IE540
L12. Multi-Armed Bandits

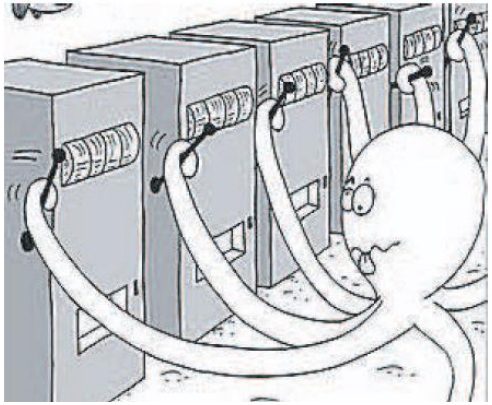
KAIST ISE
Hayong Shin

Exploration vs Exploitation Dilemma and MAB

- Online decision making : X-X dilemma is ubiquitous
  - Exploitation : make greedy decision given current information
  - Exploration : gather more information



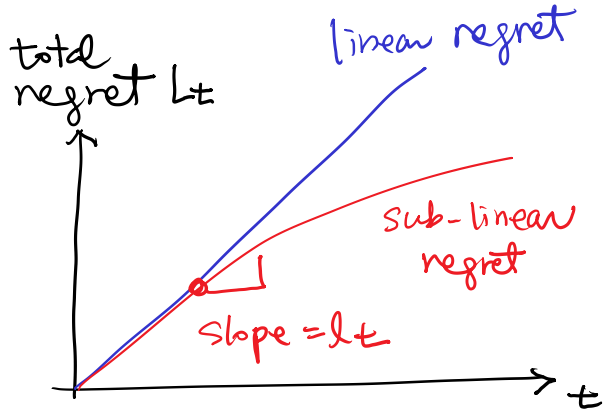
- Multi-Armed Bandit
  - Action space (arms) $\mathcal{A} = \{1,2,\dots,M\}$
    - a_t : action at time t
  - D_a : Reward distribution for action a
    - $r_t = r(a_t) \sim D_{a_t}$
  - Goal : to maximize average reward
    - usually discount is not used in MAB ($\gamma = 1$)
    - Finite horizon $R_T = \frac{1}{T}\sum_{t=1}^{T} r_t$
    - Infinite horizon $\lim_{T\to\infty} R_T$
  - MAB is single state MDP
    - single state = no notion of state



Action value and regret

- action value
  - true action value $q(a) = \mathbb{E}[r(a)] = \mathbb{E}_{r\sim D_a}[r]$
  - optimal action $a^* = \arg\max_a q(a)$
  - (optimality) gap $\Delta(a) = q(a^*) - q(a)$

- regret : measure to compare MAB algorithm performance
  - policy : π, history $h_t = (a_1, r_1, a_2, r_2, \dots, a_t, r_t) \sim \pi$
  - count $N_t(a)$: the number of selection for action a
    - $N_t(a) = \sum_{\tau=1}^{t} \mathbf{1}(a_\tau = a)$
  - one step regret
    - $l_t = \mathbb{E}_{a_t\sim\pi}[q(a^*) - q(a_t)] = \mathbb{E}_{a_t\sim\pi}[\Delta(a_t)]$
  - total regret
    - $L_t = \mathbb{E}_{h_t\sim\pi}[\sum_{\tau=1}^{t}(q(a^*) - q(a_\tau))]$
      $= \mathbb{E}_{h_t\sim\pi}[\sum_{\tau=1}^{t}\Delta(a_\tau)] = \sum_{a\in\mathcal{A}} \mathbb{E}_{h_t\sim\pi}[N_t(a)]\Delta(a)$



Policies (algorithms) for MAB to be covered

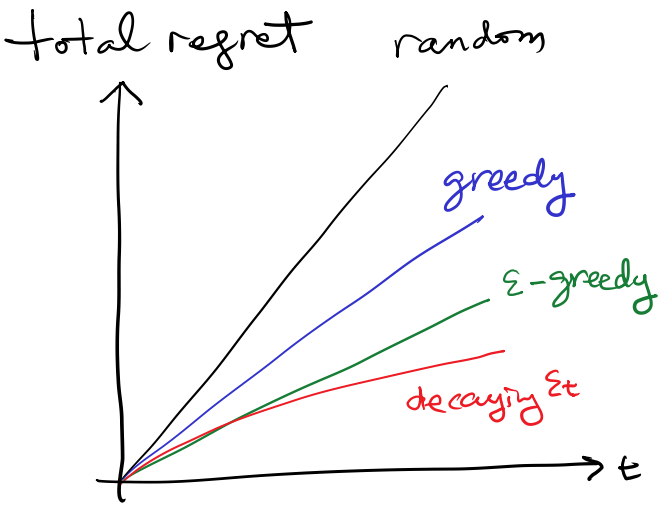
- Greedy policy : no exploration
- Naïve exploration : adding noise to greedy policy
  - random policy, ϵ-greedy, softmax, …
- Optimistic initialization
  - assume the best until proven otherwise
- Optimism in the face of uncertainty
  - select action according to optimistic guess
  - prefer actions w/ uncertain value
  - UCB
- Probability matching
  - select actions according to probability they are best
  - Thompson sampling
- Information state search (briefly)
  - Info. state MDP : an MDP w/ belief state on $q(a)$
  - solve info. state MDP using DP → Gittin Index
  - 'Knowledge gradient' : 1-step lookahead heuristic for info. state MDP
    - Frazier, Powell

Greedy policy (algorithm)

- Action value estimate (up to h_t)
  - $Q_t(a) = \frac{1}{N_t(a)} \sum_{\tau=1}^{t} r_t \mathbf{1}(a_\tau = a)$
  - Incremental update
    - $N_t(a_t) \leftarrow N_{t-1}(a_t) + 1$
    - $Q_t(a_t) \leftarrow Q_{t-1}(a_t) + \frac{1}{N_t(a_t)}\left(r_t - Q_{t-1}(a_t)\right)$
- Greedy policy (algorithm) : no exploration
  - $a_{t+1} = \arg\max_a Q_t(a)$
  - may be stuck with suboptimal choice forever
  - one step regret
    - $l_t = \mathbb{E}_{a_t \sim \pi}[\Delta(a_t)] = \sum_{a \in \mathcal{A}} P(a_t = a)\Delta(a)$
    - l_t does not decrease
  - Greedy policy has linear total regret

$Q_t(a)$	1	2	3
t \ a	1	2	3
$t=0$	20	20	20
1	10	20	20
2	10	5	20
3	10	5	9
4	12	5	9
5	11	5	9
6	10.5	5	9
7	10.7	5	9
8	10.8	5	9
	⋮		

ϵ-greedy policy (algorithm)



- Random policy (algorithm) : no exploitation
  - $a_{t+1} \sim \text{Uniform}(\mathcal{A})$
  - $P(a_t = a) = \frac{1}{|\mathcal{A}|}$, $\mathbb{E}_{h_t \sim \pi}[N_t(a)] = \frac{t}{|\mathcal{A}|}$
  - regret
    - $l_t = \mathbb{E}_{a_t \sim \pi}[\Delta(a_t)] = \frac{1}{|\mathcal{A}|}\sum_{a \in \mathcal{A}} \Delta(a)$: remains constant
    - $L_t = \frac{t}{|\mathcal{A}|}\sum_{a \in \mathcal{A}} \Delta(a)$
  - random policy has linear total regret
- ϵ-greedy policy : greedy w.p. $1-\epsilon$, random w.p. ϵ
  - $l_t = (1-\epsilon)$*greedy regret + ϵ*random regret $\geq \frac{\epsilon}{|\mathcal{A}|}\sum_{a \in \mathcal{A}} \Delta(a)$

    becomes smaller to 0
  - ϵ-greedy policy has linear total regret
- ϵ_t-greedy with decaying ϵ_t
  - can have sublinear (logarithmic) total regret
  - $\epsilon_t = \min\left\{1, \frac{c|\mathcal{A}|}{d^2 t}\right\}$, where $d = \min_{a \neq a^*} \Delta(a)$ and $c > 0$ (Auer 2002)
  - but, $\Delta(a)$ and d is unknown!!

TAQ. regret of softmax policy?

Auer et al. (2002). "Finite-time analysis of the multi-armed bandit problem", Machine Learning

Optimistic initialization

- Incremental update
  - $N_t(a_t) \leftarrow N_{t-1}(a_t) + 1$
  - $Q_t(a_t) \leftarrow Q_{t-1}(a_t) + \frac{1}{N_t(a_t)}\left(r_t - Q_{t-1}(a_t)\right)$
- Initialize $Q_0(a)$ as a large value
  - $Q_0(a) \leftarrow r_{max}$ (jack pot reward)
  - $N_0(a) = 0$ → encourage to select each arm at least once
    - $Q_1(a_1) \leftarrow r_1$ ($Q_0(a_1)$ is erased immediately)
  - $N_0(a) > 0$ → encourage early stage exploration for some time
    - Consider $Q_0(a)$ as a mean of $N_0(a)$ (hypothetical) rewards
    - Effect of $Q_0(a)$ will fade out eventually.
  - But, it does not prevent locking onto suboptimal action
    - Optimistic initialization + greedy or ϵ-greedy policy
      → linear total regret

TAO. Who is H. Robbins?

Lower bound on regret

- Difficulty of a MAB problem
  - $\Delta(a) = q(a^*) - q(a)$: penalty for wrong choice (bigger gap → bigger regret)
  - similarity in reward distribution : measured in $D_{KL}(D_a, D_{a^*})$
    - smaller $D_{KL}(D_a, D_{a^*})$ → hard to distinguish
  - difficulty = $\sum_{a \neq a^*} \frac{\Delta(a)}{D_{KL}(D_a, D_{a^*})}$

- **(Lai and Robbins Theorem)** Theoretical lower bound on total regret
  - Asymptotic total regret is at least logarithmic in number of steps

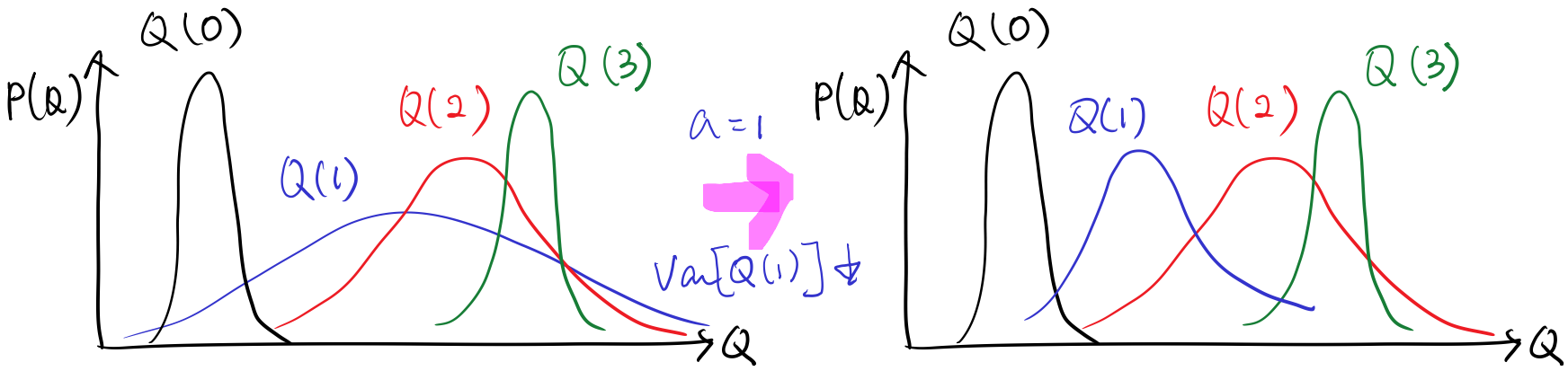
$$L_t \geq \log t \sum_{a \neq a^*} \frac{\Delta(a)}{D_{KL}(D_a, D_{a^*})} \text{ as } t \to \infty$$

  - Lower bound on L_t grows logarithmically (difficulty*$\log t$)
    - Larger L_t for difficult problem
  - Are there policies (algorithms) that can achieve this lower bound?
    - UCB, Bayesian UCB
    - Thompson sampling
    - …

TL Lai and H Robbins(1985), "Asymptotically efficient adaptive allocation rules", Advances in Applied Mathematics 6, 1, 4–22.

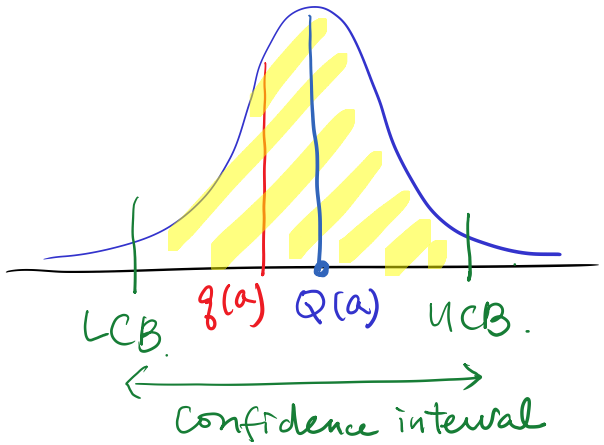
Optimism in the face of uncertainty

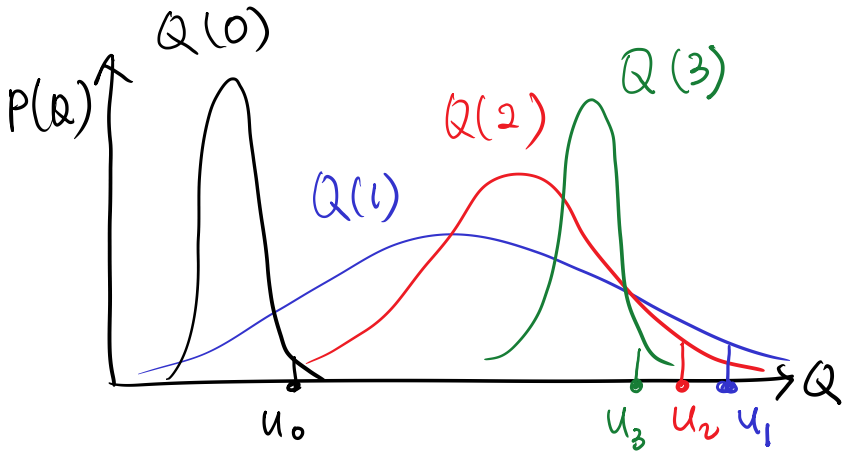
- "Optimism in the face of uncertainty" principle
  - select action based on optimistic guess if uncertain
    - the more uncertain about a, the more exploration of a is necessary
  - which action would you take?
    - action 0: no reason to take this action
    - action 3: exploitation
    - action 1: exploration
  - after taking $a = 1$, posterior of $Q(1)$ changes
  - how can we make this as a policy?



UCB(Upper Confidence Bound) policy

- Estimate confidence bound $U_t(a)$ for each arm a
  - s.t. $\mathrm{P}[q(a) > Q_t(a) + U_t(a)] \leq p$ $(1-p$: confidence level)
- Take action $a_{t+1} = \arg\max_a Q_t(a) + U_t(a)$
  - $Q_t(a)$ for exploitation
  - $U_t(a)$ for exploration
- How to determine $U_t(a)$
  - based on $N_t(a)$ and t only → UCB1
    - $N_t(a)$ is large → more observations on a → small $U_t(a)$
  - based on posterior of $r(a)$ → Bayesian UCB





UCB1 (Auer 2002)

- Hoeffding's inequality
  - $X_1, X_2, \ldots, X_n$: i.i.d. random variable in $[0,1]$.
  - $\bar{X}_n = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$
  - ➔ $\mathrm{P}[\mathbb{E}[X] > \bar{X}_n + u] < e^{-2nu^2}$

- Replacing $\mathbb{E}[X] \to q(a), \bar{X}_t \to Q_t(a), u \to U_t(a), n \to N_t(a)$
  - $\mathrm{P}[q(a) > Q_t(a) + U_t(a)\ ] \leq e^{-2N_t(a)U_t(a)^2} = p_t$
  - $e^{-2N_t(a)U_t(a)^2} = p_t$ ➔ $U_t(a) = \sqrt{-\frac{\log p_t}{2N_t(a)}}$
  - if we choose $p_t = t^{-4}$ ➔ $U_t(a) = \sqrt{\frac{2\log t}{N_t(a)}}$

TAQ. What if we use $p_t = t^{-\alpha}$?

- UCB1 policy: $a_{t+1} = \arg\max_a Q_t(a) + \sqrt{\frac{2\log t}{N_t(a)}}$
  - UCB1 achieves logarithmic asymptotic total regret
  - $L_t \leq 8\log t \sum_{a \neq a^*} \frac{1}{\Delta(a)} + \left(1 + \frac{\pi^2}{3}\right)\sum_a \Delta(a) = O(\log t)$

Auer et al. (2002). "Finite-time analysis of the multi-armed bandit problem", Machine Learning

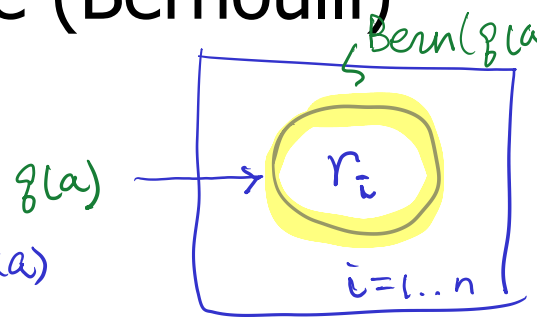
Interpretation and criticism of UCB1

- UCB1 policy: $a_{t+1} = \arg\max_a \{Q_t(a) + U_t(a)\},\ U_t(a) = \sqrt{\frac{2\log t}{N_t(a)}}$
  - $U_t(a) \propto \sqrt{\frac{1}{N_t(a)}}$: this is typical $\left(\text{var}[\bar{X}_n] = \frac{\text{var}[X_i]}{n}\right)$
    - $N_t(a)$ large → less uncertainty in $Q_t(a)$ → smaller $U_t(a)$
  - $U_t(a) \propto \sqrt{2\log t}$: this come from setting $p_t = t^{-4}$ (strange choice?)
    - this is required to prove the UCB1 achieves Lai-Robbins regret lower bound
    - Note: decrease p_t → increase confidence level $1 - p_t$ → more exploration
  - $U_t(a)$ is based on that reward distribution is bounded in [0,1]
    - Bernoulli bandit satisfies this range
      - even in this case, modifying $U_t(a)$ with sample std. dev. works better
      - UCB1-Tuned
    - If each arm has different range $[l_a, u_a]$ → UCB1 can be easily modified
    - If reward distribution is normal : UCB1-Normal (see A2)
      - UCB1 modified for normally distributed reward
      - $U_t(a) = 4\sigma_t(a)\sqrt{\frac{\log t}{N_t(a)}}$, where $\sigma_t(a)$ is sample std. dev. of reward from a
    - see Auer 2002 for the details
    - TAQ. Can we use this when reward distribution is not normal?

Auer et al. (2002). "Finite-time analysis of the multi-armed bandit problem", Machine Learning

Bayesian posterior update (Bernoulli)

- Frequentist setting
  - $r(a) \sim \mathrm{Bern}(q(a))$ $= \begin{cases} 1 & \text{w.p. } q(a) \\ 0 & \text{otherwise} \end{cases}$

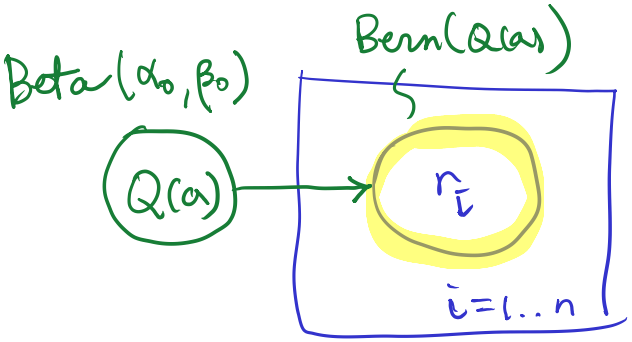


$Q(a) = \frac{1}{n} \sum_i r_i$

$h_n = \{r_1, \cdots, r_n\}$

- Bayesian setting
  - $r(a) \sim \mathrm{Bern}(Q(a)|h_t)$

Prior of Q(a)
+ Observation (data)
⇒ Posterior of $Q(a)|h_n$



Posterior
$Q(a)|h_n \sim \mathrm{Beta}(\alpha_n, \beta_n)$
$= Q_n(a)$

$\alpha_n = \alpha_0 + \sum_i r_i$
$\beta_n = \beta_0 + n - \sum_i r_i$

$Q_0(a)$ Data $Q(a)|h_1$ $Q(a)|h_2$

$\mathrm{Beta}(\alpha_0, \beta_0) \to r_1 \to \mathrm{Beta}(\alpha_1, \beta_1) \to r_2 \to \mathrm{Beta}(\alpha_2, \beta_2) \to \cdots$

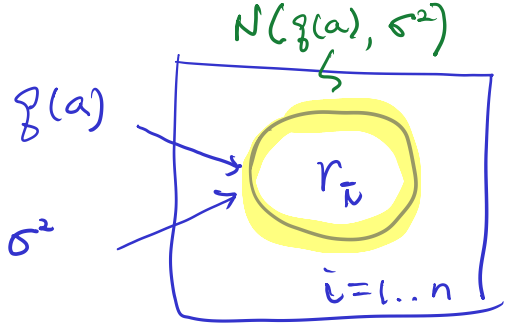
$\alpha_1 = \alpha_0 + r_1$
$\beta_1 = \beta_0 + 1 - r_1$

$\alpha_2 = \alpha_1 + r_2$
$\beta_2 = \beta_1 + 1 - r_2$

$\alpha_n = \alpha_{n-1} + r_n$
$\beta_n = \beta_{n-1} + 1 - r_n$

Bayesian posterior update (Normal)

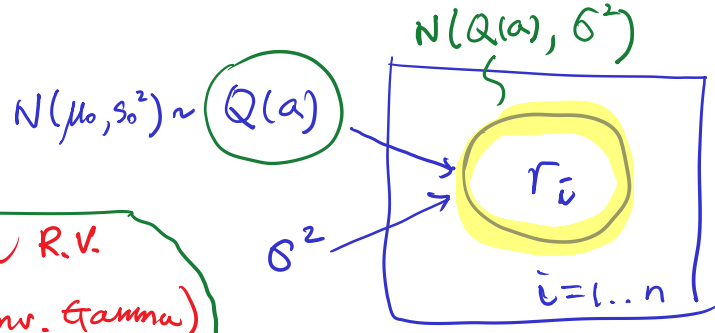
- Frequentist setting
  - $r_i \sim N(q(a), \sigma(a)^2)$
- Bayesian setting
  - $r_i \sim N(Q(a), \sigma(a)^2)$



$$Q(a) = \frac{1}{n}\sum_i r_i$$

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_i (r_i - Q(a))^2$$

$$h_n = \{ r(a)^{(1)}, \dots, r(a)^{(n)} \}$$



Posterior

$$Q(a)|h_n \sim N(\mu_n, s_n^2)$$

$$Q_n(a)$$

$\sigma^2(a)$ can also be modeled as R.V.

$\Rightarrow \sigma^2(a) \sim IG(\alpha_0, \beta_0)$ (Inv. Gamma)

$$\begin{cases} \mu_n = \dfrac{s_0^2 \sum r_i + n\sigma^2 \mu_0}{s_0^2 + n\sigma^2} \\ \dfrac{1}{s_n^2} = \dfrac{1}{s_0^2} + \dfrac{n}{\sigma^2} \end{cases}$$

$Q_0(a)$ Data update $Q(a|h_1) = Q_1(a)$

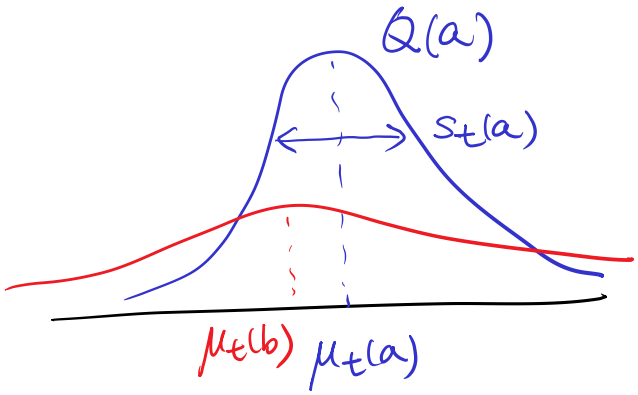
$N(\mu_0, s_0^2) \rightarrow r_1 \longrightarrow N(\mu_1, s_1^2) \rightarrow r_2 \dots$

$$\begin{cases} \mu_n = \dfrac{s_{n-1}^2 r_n + \sigma^2 \mu_{n-1}}{s_{n-1}^2 + \sigma^2} \\ \dfrac{1}{s_n^2} = \dfrac{1}{s_{n-1}^2} + \dfrac{1}{\sigma^2} \end{cases}$$

$$\mu_n = \frac{\frac{1}{\sigma^2} r_n + \frac{1}{s_{n-1}^2}\mu_{n-1}}{\frac{1}{\sigma^2} + \frac{1}{s_{n-1}^2}}$$

Bayesian bandit – independent normal

- history (observation data)
  - $h_t = (a_1, r_1, a_2, r_2, \dots, a_t, r_t)$
- Observation model
  - $r(a) \sim N(Q(a), \sigma^2(a))$
- prior
  - $Q_0(a) = Q(a) | h_0 \sim N(\mu_0(a), s_0^2(a))$



- posterior
  - $Q_t(a) = Q(a) | h_t \sim N(\mu_t(a), s_t^2(a))$

TAQ. Don't we need $\sqrt{\frac{1}{N_t(a)}}$ here?

- policy
  - Bayesian UCB : $a_{t+1} = \arg\max_a \{\mu_t(a) + c \cdot s_t(a)\}$
    - c : level of exploration
  - Probability matching : Thompson sampling

TAQ. How about using $\sqrt{\log t}$ here?

Thompson sampling

- probability matching
  - select action a w.p. a is the optimal action
    - according to the posterior distribution $Q(a)|h_t$
  - stochastic policy
    - $\pi(a|h_t) = \mathrm{P}[Q(a) > Q(a'), \forall a' \neq a|h_t] = \mathrm{P}[Q_t(a) > Q_t(a'), \forall a' \neq a]$
- Thompson sampling
  - simple way of implementing probability matching
    - very old idea (1930's)
    - use sample from posterior
    - achieves Lai-Robbins lower bound of regret
  - procedure (for normal case)
    - given : posterior $Q(a)|h_t \sim N\left(\mu_t(a), s_t^2(a)\right)$
    - sample $\tilde{Q}(a)$ from $N\left(\mu_t(a), s_t^2(a)\right)$ for $\forall a$
    - take action $a_{t+1} = \arg\max_a \tilde{Q}(a)$, observe reward r_{t+1}
      - $\mathbb{E}\left[r(a)|\tilde{Q}(a)\right] = \tilde{Q}(a)$ in this case. Otherwise, compute & compare $\mathbb{E}\left[r(a)|\tilde{Q}(a)\right]$
    - update posterior $Q(a_{t+1})|h_{t+1}$ with r_{t+1}

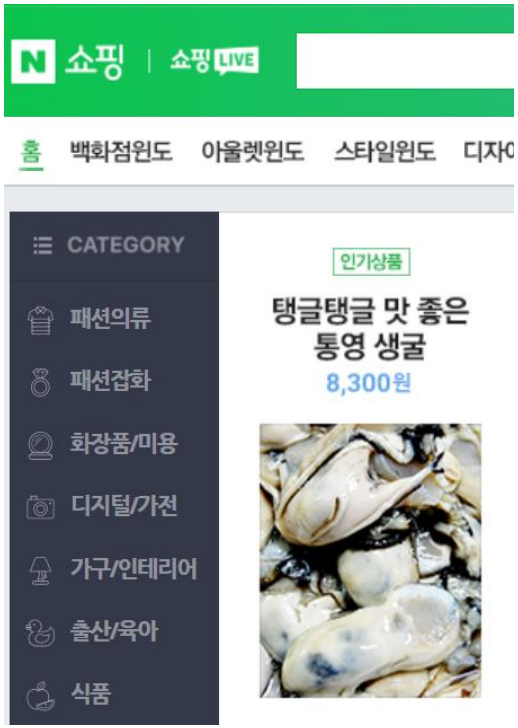
$Q_t(a) = Q(a)|h_t$: posterior r.v.
$p[q|h_t]$: posterior pdf
$Q_t(a) \sim p[q|h_t]$

TAQ. If we use k>1 samples for each arm and take sample average, would it perform better? Can we achieve Lai-Robbins? What would happen if k→∞?

Information state search

- Consider an MDP
  - state : information state = knowledge state = belief state
    - information state : posterior distribution parameters
    - $Q(a)|h_t \sim Beta(\alpha_t(a), \beta_t(a))$
  - action : arm a to pull
  - state transition model
    - observation: $r(a) \sim \text{Bern}(Q(a)|h_t)$
    - posterior update
      - $\alpha_{t+1}(a) = \alpha_t(a) + r(a)$, $\beta_{t+1}(a) = \beta(a) + 1 - r(a)$
  - reward : $r(a)$
- Gittins index : solution to this MDP with finite horizon
  - DP solution requires heavy computation
- Knowledge gradient
  - 1-step lookahead heuristic to this MDP
  - P. Frazier (2009, PhD thesis), "Knowledge-gradient methods for statistical learning"
  - Frazier, Powell, Dayanik, "Knowledge-gradient policy for sequential information collection", SIAM J. on Control & Optimization (2008)

Further studies : Contextual bandit



- Contextual bandit
  - action $\mathcal{A} = \{1,2,\dots,M\}$: which ad to show?
  - context c : user of website
    - context set $\mathcal{C} = \{1,2,\dots,K\}$: can be extremely large
    - similar to state, but $p(c'|c,a) = p(c')$
  - reward : $r_t = r(c_t, a_t) = \mathbf{1}(click\ ad)$
    - reward varies from one user to another
    - $q(c,a) = \mathbb{E}[r|c,a]$
  - Naïve approach : solve MAB for each context c
    - prohibitively inefficient
    - similar context (user) will show similar reward function
- LinUCB : construct linearly parameterized value function
  - $Q_\theta(c,a) = \phi(c,a)^T\theta$
    - $\phi(c,a)$: feature vector of context c and action a (assumed to be known)
    - need to update θ while collecting more data → X-X
  - apply UCB on parameter space $\theta \in \Theta$ (see A3)
- For more details, refer to the paper
  - Li, Lihong, et al. "A contextual-bandit approach to personalized news article recommendation." *Proceedings of the 19th international conference on World wide web*. 2010.

Exploration in MDP

- Many of the ideas in MAB can also applied to MDP
  - optimistic initialization
  - optimism in the face of uncertainty principle, UCB
  - Bayesian RL : Bayesian UCB, Thompson sampling
- Exploration in MDP is different from MAB
  - reward is not delayed in MAB
    - easier to estimate value function in MAB
    - value function estimation is harder in MDP than MAB
  - no (one) state in MAB vs. huge state space in MDP
    - requires huge memory and long time for maintaining MAB information for every state
    - How to combine MAB with function approximation
      - how to quantify uncertainty of value function estimate
      - DQN + UCB?
      - PPO + Thompson sampling?

A1. UCB(α) policy

- UCB(α) policy: $a_{t+1} = \arg\max_a \left\{ Q_t(a) + \sqrt{\frac{\alpha \log t}{2N_t(a)}} \right\}$
- Regret upper bound for UCB(α)
  - $L_t \leq 2\alpha \log t \sum_{a \neq a^*} \frac{1}{\Delta(a)} + \frac{\alpha}{\alpha - 2} \sum_a \Delta(a) = O(\log t)$ if $\alpha > 2$
    - $L_t = \sum_a \Delta(a)\, \mathbb{E}[N_t(a)]$
    - $\mathbb{E}[N_t(a)] \leq \frac{2\alpha \log t}{\Delta(a)^2} + \frac{\alpha}{\alpha - 2}$
    - see lecture note of D. Katselis for UIUC, ECE586 MDP & RL
    - There, it is also mentioned that UCB(α) with $\alpha > 1$ can also have $O(\log t)$ regret upper bound

A2. UCB1-Normal (Auer 2002)

- Hoeffding's inequality for Gaussian (Kang 2020)
  - $X_1, X_2, \ldots, X_n \sim$ i.i.d. $N(\mu, \sigma^2)$.
  - $\bar{X}_n = \frac{1}{n}(X_1 + X_2 + \cdots + X_n) \sim N(\mu, \sigma^2/n)$

  ➔ $\mathrm{P}[\mathbb{E}[X] > \bar{X}_n + u] < e^{-\frac{nu^2}{2\sigma^2}}$

- Replacing $\mathbb{E}[X] \to q(a), \bar{X}_t \to Q_t(a), u \to U_t(a), n \to N_t(a), \sigma^2 \to \sigma^2(a)$
  - $\mathrm{P}[q(a) > Q_t(a) + U_t(a)\ ] \le e^{-\frac{N_t(a)U_t(a)^2}{2\sigma^2(a)}} = p_t$
  - $e^{-\frac{N_t(a)U_t(a)^2}{2\sigma^2(a)}} = p_t$ ➔ $U_t(a) = \sigma(a)\sqrt{-\frac{2\log p_t}{N_t(a)}}$
  - if we choose $p_t = t^{-4}$ ➔ $U_t(a) = 2\sqrt{2}\sigma(a)\sqrt{\frac{\log t}{N_t(a)}}$
  - If $\sigma^2(a)$ is not known and sample variance $\sigma_t^2(a)$ is used instead, some correction is required. (Proof gets more complicated.)
  - $U_t(a) = 4\sigma_t(a)\sqrt{\frac{\log t}{N_t(a)}}$

Auer et al. (2002). "Finite-time analysis of the multi-armed bandit problem", Machine Learning
W.Kang and S.Yun (2020), "Lecture note on Concentration Inequalities", KAIST

A3. Lin UCB

$$Q_\theta(s,a) = \phi(s,a)^T \theta$$

$$h_t = \{(s_\tau, a_\tau, r_\tau) ; \tau = 0..t-1\}$$

$$\phi(s_\tau,a_\tau)^T\theta = r_\tau + \varepsilon_\tau \quad \Rightarrow \quad L(\theta) = \frac{1}{t}\sum_{\tau=0}^{t-1} \varepsilon_\tau^2 = \frac{1}{t}\sum_{\tau=0}^{t-1}\left(\phi(s_\tau,a_\tau) - r_\tau\right)^2$$

$$\min_\theta L(\theta) \Rightarrow \theta_t = A_t^{-1} b_t, \qquad A_t = \sum_{\tau=0}^{t-1}\phi(s_\tau,a_\tau)\,\phi(s_\tau,a_\tau)^T, \quad b_t = \sum_{\tau=0}^{t}\phi(s_\tau,a_\tau)\,r_\tau$$

$Q_{\theta_t}(s,a) = \phi(s,a)^T A_t^{-1} b_t$ ⟶ how good is a, given context s ?

$Var(Q_{\theta_t}(s,a)) = \phi(s,a)^T A_t^{-1} \phi(s,a) = \sigma^2_{\theta_t}(s,a)$ → how uncertain is $Q_{\theta_t}(s,a)$?

$UCB = Q_{\theta_t}(s,a) + c\,\sigma_{\theta_t}(s,a)$

⇒ Select a_t at context s_t

s.t. $a_t = \underset{a}{\operatorname{argmax}}\left[Q_{\theta_t}(s_t,a) + c\,\sigma_{\theta_t}(s_t,a)\right]$